Q1. What is an I²S controller? What are its features and applications?

Srinivas Salloori
Through e-mail

A1. I²S stands for integrated inter-chip sound. It is an electrical serial bus interface standard used for connecting digital audio devices. The I²S audio interface provides a bidirectional, synchronous, serial interface to off-chip audio devices. It complies with the inter-IC sound (I²S) bus specification from Philips Semiconductor, and is used for integrating audio transmission channels into an application-specific integrated circuit (ASIC) or a system-on-a-chip (SoC) design.

I²S is a well-known stereo audio transmission standard, widely used for connecting system elements such as analogue-to-digital and digital-to-analogue converters. By using the I²S-AHB (advance high-performance bus), you can easily integrate the core used for multichannel audio transmission with the master system based on a standard AHB bus for further data processing. The bus supports master-slave and receiver-transmitter modes. I²S signals can easily be transferred via Ethernet-specified connection hardware (8P8C plugs, jacks, Cat-5e cabling). It is commonly used for carrying PCM information between the CD transport and the DAC in a CD player.

Q2. I am an electronics and telecom engineer, presently teaching at an engineering college in Goa. I want to become a ham radio operator. What's the procedure?

Aarti Kamat
Through e-mail

A2. Ham radio operators belong to a community that uses radio transmitters and receivers to communicate with the members of the community. To become a ham radio operator, you need a transceiver and a licence for operating/possessing a radio. For obtaining the licence, you have to qualify the examination conducted by the Ministry of Communications, government of India. Wireless monitoring stations, department of telecommunication (DoT) under the Ministry of Communications, conduct these tests.

Any individual above the age of twelve is permitted to appear for amateur station operator licence examination. No educational qualification is prescribed. It takes two months of training to become eligible for the examination. The examination is conducted in three subjects: Morse code (transmission and reception), communication procedure and basic electronics. The licences are issued by Wireless Planning and Coordination Wing of DOT, after you pass the test in any of the following grades:

1. Restricted Grade II: Permitting use of VHF/UHF only (walkie-talkies)

2. Grade II: Permitting HF/VHF/UHF frequencies but with limited transmitting power

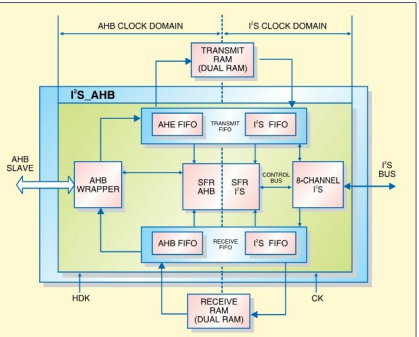
3. Grade I: Permitting all amateur frequencies with higher power including the latest techniques

4. Advance Grade: Permitting higher power and advanced techniques including satellite communication.

Many books are available for the beginners. One of them is an EFY publication, *ABC of Amateur Radio and Citizen Band*, authored by Rajesh Verma. The book costs Rs 75 and can be purchased from EFY associates Kits'n'Spares. For more information, refer to www.hamradioindia.org

Q3. What is a static converter? What types of static converters are available? Is the AC/DC power supply a static converter?

Prakash
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Functional description
The main blocks of I²S-AHB are:
- I²S: Multi-channel I²S audio core
- AHB wrapper: Bidirectional wrapper from FIFOs and SFR block to AMBA AHB slave interface
- SFR: Block of special function registers
- AHB FIFO: AHB interface/controller of the transmit/receive RAM
- I²S FIFO: I²S interface/controller for the transmit/receive RAM interfaces

Interfaces
There are two interfaces in the I²S core:
- Serial interface: For transfer of information between devices connected to the bus
- AHB slave interface: Allows access to all features of the I²S controller

Block diagram of CAST- I²S-AHB